New Wheels Project

Introduction to SQL

Problem Statement

Business Context

A lot of people in the world share a common desire: to own a vehicle. A car or an automobile is seen as an object that gives the freedom of mobility. Many now prefer pre-owned vehicles because they come at an affordable cost, but at the same time, they are also concerned about whether the after-sales service provided by the resale vendors is as good as the care you may get from the actual manufacturers.

New-Wheels, a vehicle resale company, has launched an app with an end-to-end service from listing the vehicle on the platform to shipping it to the customer's location. This app also captures the overall after-sales feedback given by the customer.

Objective

New-Wheels sales have been dipping steadily in the past year, and due to the critical customer feedback and ratings online, there has been a drop in new customers every quarter, which is concerning to the business. The CEO of the company now wants a quarterly report with all the key metrics sent to him so he can assess the health of the business and make the necessary decisions.

As a data analyst, you see that there is an array of questions that are being asked at the leadership level that need to be answered using data. Import the dump file that contains various tables that are present in the database. Use the data to answer the questions posed and create a quarterly business report for the CEO.

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Business Questions

Question 1: Find the total number of customers who have placed orders. What is the distribution of the customers across states?

```
SELECT
   state,
   COUNT(DISTINCT customer_id) AS total_customers
FROM
   customer_t
GROUP BY state
ORDER BY total_customers DESC;
```

Output:

Showing first 10 rows out of 49 rows

state	total_customers
Texas	97
California	97
Florida	86
New York	69
District of Columbia	35
Ohio	33
Colorado	33
Alabama	29
Washington	28

Result: Passed

Observations and Insights:

- Texas and California lead with 97 customers each and represent a disproportionate share of the base. Marketing and inventory should align with that reality.
- Customer concentration is heavily skewed toward large coastal and sunbelt states.
- After the top four, there is a steep drop off. States from DC through Arizona range between 26 and 35 customers, about three to four times fewer than the leaders. That long tail signals meaningful growth opportunities in underpenetrated mid tier markets.

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Question 2: Which are the top 5 vehicle makers preferred by the customers?

Solution Query:

```
SELECT

   vehicle_maker,

   COUNT(DISTINCT customer_id) AS total_customers

FROM product_t p

JOIN order_t o ON p.product_id = o.product_id

GROUP BY vehicle_maker

ORDER BY total_customers DESC

LIMIT 5;
```

Output:

Showing 5 rows

vehicle_maker	total_customers
Chevrolet	83
Ford	63
Toyota	52
Pontiac	50
Dodge	50

Result: Passed

Observations and Insights:

- Chevrolet leads with 83 customers, about 32% ahead of Ford at 63. It is the clear market leader, which likely points to strong brand pull or a pricing advantage.
- American brands take four of the top five spots. Chevy, Ford, Pontiac, and Dodge dominate, with Toyota as the only foreign brand in the top group. That suggests a strong domestic preference within the customer base.
- From 3rd to 5th, it is tight. Toyota, Pontiac, and Dodge are all clustered between 50 and 52 customers. Small changes in pricing, incentives, or inventory could easily reshuffle those rankings, so it is a segment worth watching closely.

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Question 3: Which is the most preferred vehicle maker in each state?

Solution Query:

```
SELECT state,

     vehicle_maker,

     total_cust

FROM

(SELECT c. state,

     p.vehicle_maker,

     COUNT(o.customer_id) as total_cust,

     RANK() over(PARTITION BY c.state ORDER BY COUNT(o.customer_id) DESC) as rnk

FROM order_t o

JOIN product_t p

ON p.product_id =o.product_id




JOIN customer_t c

ON o.customer_id = c.customer_id

GROUP BY 1,2

ORDER BY 1, 3 DESC) as a

WHERE rnk = 1 ;
```

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Output:

state	vehicle_maker	total_cust
Alabama	Dodge	5
Alaska	Chevrolet	2
Arizona	Pontiac	3
Arizona	Cadillac	3
Arkansas	Volkswagen	1
Arkansas	Suzuki	1
Arkansas	Pontiac	1
Arkansas	Mitsubishi	1
Arkansas	GMC	1
Arkansas	Chevrolet	1

Result: Passed

Observations and Insights:

- Low-volume states are basically noise. Arizona is tied 3–3 between Pontiac and Cadillac. Arkansas has six brands with one customer each. There's no real dominance there, so strong conclusions would be unreliable.
- National brand leadership doesn't hold at the state level. Chevrolet is #1 overall but only leads Alaska here. Dodge leads Alabama and Arizona is tied. Brand loyalty clearly varies by region, so a strategy built purely on national rankings will miss in certain markets.
- Arkansas is underpenetrated and wide open. Six brands with one customer each means no one owns the market. It's low volume, but it's also one of the easiest places to establish share with a focused campaign compared to saturated states like Texas or California.

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Question 4: Find the overall average rating given by the customers. What is the average rating in each quarter?

Consider the following mapping for ratings: "Very Bad": 1, "Bad": 2, "Okay": 3, "Good": 4, "Very Good": 5

Solution Query:

```
SELECT quarter_number,

     ROUND(AVG(feedback_num),2) as avg_rating

FROM

(SELECT quarter_number,

     CASE WHEN customer_feedback = 'Very Bad' then 1

          WHEN customer_feedback = 'Bad' then 2

          WHEN customer_feedback = 'Okay' then 3

          WHEN customer_feedback = 'Good' then 4

          WHEN customer_feedback = 'Very Good' then 5

      END AS feedback_num

FROM order_t) as a

GROUP BY 1

ORDER BY 1 ;

SELECT ROUND(AVG(feedback_num),2) as avg_rating

FROM

(SELECT quarter_number,

  CASE WHEN customer_feedback = 'Very Bad' then 1

          WHEN customer_feedback = 'Bad' then 2

          WHEN customer_feedback = 'Okay' then 3

          WHEN customer_feedback = 'Good' then 4

          WHEN customer_feedback = 'Very Good' then 5
```

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

```
      END AS feedback_num

FROM order_t) as a ;
```

Output:

Result: Passed

Showing 4 rows

quarter_number	avg_rating
1	3.55
2	3.35
3	2.96
4	2.4

Showing 1 rows

avg_rating
3.14

Observations and Insights:

- Ratings decline every quarter from 3.55 in Q1 to 2.40 in Q4. That is a 1.15 point drop in a single year. A 2.40 average means the typical Q4 experience is closer to bad than okay. That is a real problem, not normal fluctuation.
- Q3 is the turning point. At 2.96, it is the first time the average falls below 3.0, which is effectively the line for acceptable. That is when sentiment shifts negative, and it lines up directly with shipping jumping to 117 days. The correlation is hard to ignore.
- The 3.14 full year average is misleading. It is propped up by Q1 and Q2. The second half averages around 2.68, which tells the more honest story. If nothing changes, dropping below a 2.0 average next year is a realistic risk.

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Question 5: Find the percentage distribution of feedback from the customers. Are customers getting more dissatisfied over time?

Solution Query:

```
SELECT

   quarter_number,

   ROUND(SUM(CASE WHEN customer_feedback = 'Very Bad'  THEN 1 ELSE 0 END) * 100.0 /
COUNT(*), 2) AS very_bad_pct,

   ROUND(SUM(CASE WHEN customer_feedback = 'Bad'       THEN 1 ELSE 0 END) * 100.0 /
COUNT(*), 2) AS bad_pct,

   ROUND(SUM(CASE WHEN customer_feedback = 'Okay'      THEN 1 ELSE 0 END) * 100.0 /
COUNT(*), 2) AS okay_pct,

   ROUND(SUM(CASE WHEN customer_feedback = 'Good'      THEN 1 ELSE 0 END) * 100.0 /
COUNT(*), 2) AS good_pct,

   ROUND(SUM(CASE WHEN customer_feedback = 'Very Good' THEN 1 ELSE 0 END) * 100.0 /
COUNT(*), 2) AS very_good_pct

FROM order_t

GROUP BY quarter_number

ORDER BY quarter_number;
```

Output:

Result: Passed

Showing 4 rows

quarter_number	very_bad_pct	bad_pct	okay_pct	good_pct	very_good_pct
1	10.97	11.29	19.03	28.71	30
2	14.89	14.12	20.23	22.14	28.63
3	17.9	22.71	21.83	20.96	16.59
4	30.65	29.15	20.1	10.05	10.05

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Observations and Insights:

- Sentiment reverses by Q4. “Very Good” falls from 30% to 10%, while “Very Bad” rises from 10% to 30%. By Q4, 59% of feedback is negative and dissatisfaction clearly takes over in the second half.
- Q3 is the inflection point. Negative sentiment at 39% first exceeds positive at 36%. That is likely when operational strain or execution issues begin impacting customers.
- “Okay” stays flat around 19 to 21%. The shift is not in neutral customers. It is satisfied customers moving into dissatisfied categories, which signals a systemic decline in experience quality.

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Question 6: What is the trend of the number of orders by quarter?

Solution Query:

```
SELECT

   quarter_number,

   COUNT(order_id) AS total_orders

FROM order_t

GROUP BY quarter_number

ORDER BY quarter_number;
```

Output:

Showing 4 rows

quarter_number	total_orders
1	310
2	262
3	229
4	199

Result: Passed

Observations and Insights:

- Orders decline every quarter, from 310 in Q1 to 199 in Q4. That is a 36% drop across the year. This looks like it might not be seasonal noise, it's a sustained downward trend, and paired with worsening feedback, it suggests customer experience issues are driving churn.
- The largest single drop happens from Q1 to Q2, down 48 orders. The slowdown starts early, well before sentiment fully collapses in Q4.
- Q4 closes at 199 orders, the weakest point of the year. That lines up with 59% negative feedback, reinforcing the likelihood of a feedback to churn cycle. Fixing customer experience needs to be a priority going into next year.

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Question 7: Calculate the net revenue generated by the company. What is the quarter-over-quarter % change in net revenue?

Solution Query:

```
SELECT quarter_number,revenue,

     100* (Revenue - LAG(revenue,1) OVER(ORDER BY quarter_number))/

     LAG(revenue,1) OVER(ORDER BY quarter_number) as qoq

FROM

(SELECT quarter_number,

     SUM(quantity *( vehicle_price- ((discount/100)* vehicle_price))) as revenue

FROM order_t

GROUP BY 1 ) as a;
```

Output:

Showing 4 rows

quarter_number	revenue	qoq
1	39421580.15929598	
2	32715830.33996199	-17.010352685603124
3	29229896.19364898	-10.655190805458437
4	23346779.63060599	-20.127052535757088

Result: Passed

Observations and Insights:

- Revenue declined every quarter with no recovery. Down 17 percent in Q2, 10.66 percent in Q3, and 20.13 percent in Q4. That is roughly a 41 percent drop from Q1 to Q4, about 16 million in lost revenue.
- Q4 is the biggest concern. The decline nearly doubles from Q3, showing acceleration, not stabilization.
- The warning sign was Q2. That 17 percent drop should have triggered action. Shipping delays also started climbing then, which aligns directly with the revenue slide.

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Question 8: What is the trend of net revenue and orders by quarters?

Solution Query:

```
SELECT

   o.quarter_number,

   COUNT(o.order_id) AS total_orders,

   ROUND(SUM(o.quantity * p.vehicle_price * (1 - o.discount/100)), 2) AS net_revenue

FROM order_t o

JOIN product_t p ON o.product_id = p.product_id

GROUP BY o.quarter_number

ORDER BY o.quarter_number;
```

Output:

Showing 4 rows

quarter_number	total_orders	net_revenue
1	310	39421580.16
2	262	32715830.39
3	229	29229896.19
4	199	23346779.51

Result: Passed

Observations and Insights:

- Revenue and orders decline together every quarter. Orders drop steadily, and revenue follows the same pattern, falling from 39.4 million in Q1 to 23.3 million in Q4. That is roughly a $16M or 41% decline across the year. Volume loss is clearly driving revenue loss.
- Revenue falls faster than orders in the second half. Orders are down 36% from Q1 to Q4, but revenue is down 41%. That suggests lower revenue per order, potentially from heavier discounting or weaker pricing power in Q3 and Q4.
- Q4 is the low point on both fronts, with 199 orders and 23.3 million in revenue. Layer that with 59% negative feedback, and the pattern is clear. Fewer orders, lower revenue, worsening sentiment. It is reinforcing a negative cycle that needs direct intervention.

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Question 9: What is the average discount offered for different types of credit cards?

```
SELECT

   credit_card_type,

   ROUND(AVG(discount), 2) AS avg_discount

FROM order_t o

JOIN customer_t c ON o.customer_id = c.customer_id

GROUP BY credit_card_type

ORDER BY avg_discount DESC;
```

Output:

Showing first 10 rows out of 16 rows

credit_card_type	avg_discount
laser	0.64
mastercard	0.63
visa-electron	0.62
maestro	0.62
instapayment	0.62
china-unionpay	0.62
americanexpress	0.62
switch	0.61
jcb	0.61
diners-club-us-ca	0.61

Result: Passed

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Observations and Insights:

- Discounts are almost identical across all card types. The range is just .61 to .64 percent. That suggests discounting is applied uniformly, not strategically differentiated by card type or partnership.
- Laser averages the highest discount at .64 percent, followed closely by Mastercard at .63 percent. Visa Electron, Amex, and China UnionPay sit in the middle. The slight lift for Laser likely reflects a small, highly discounted subset rather than a broader strategy.
- There are 16 different card types in the dataset, which shows a diverse payment mix. Given the near identical discount rates, there may be room to implement tiered discounts tied to preferred payment partners to better manage and influence behavior.

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Question 10: What is the average time taken to ship the placed orders for each quarter?

Solution Query:

```
SELECT

   quarter_number,

   ROUND(AVG(julianday(ship_date) - julianday(order_date)), 2) AS avg_shipping_days

FROM order_t

GROUP BY quarter_number

ORDER BY quarter_number;
```

Output:

Showing 4 rows

quarter_number	avg_shipping_days
1	57.17
2	71.11
3	117.76
4	174.1

Result: Passed

Observations and Insights:

- Shipping times worsen sharply over the year. Average delivery time jumps from 57 days in Q1 to 174 days in Q4, more than tripling. That kind of delay almost certainly drives the spike in negative feedback.
- The biggest shift happens in Q3, where shipping time increases from 71 days to about 118 days in one quarter. That points to a major operational issue midyear, likely supply chain disruption, logistics problems, or inventory constraints.
- The timing lines up with the drops in orders, revenue, and sentiment. As wait times stretch to nearly six months, dissatisfaction rises and demand falls. Improving logistics should be the primary focus if the goal is to reverse the downward trend.

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Business Metrics Overview

Total Revenue	Total Orders	Total Customers	Average Rating
$124,714,086.26	1000	994	3.14
Last Quarter Revenue	**Last quarter Orders**	**Average Days to Ship**	**% Good Feedback**
$23,346,779.51	199	97.96	44.1

Note: These values must be derived using SQL queries. Some of them may have already been obtained while answering previous questions.

Business Recommendations

- **Fix shipping and logistics first.** Shipping times went from 57 days to 174 days in Q4 and that's the biggest problem that is impacting everything else. They need to audit the supply chain, negotiate carrier terms, and get back under a 60 day target. With positive feedback at only 44% and negative feedback at 59% in Q4, customer trust is clearly being damaged by fulfillment issues.
- **Focus on retention in Texas, California, and Florida.** These three states represent the largest concentration of customers (280 total) and the best opportunity to recover revenue. Orders dropped to just 199 in Q4 across the entire company so there's room to win back volume. A targeted outreach campaign with incentives for repeat purchases in those markets could help stabilize demand before things decline further.
- **Make discounts strategic instead of flat.** Chevy and Ford drive most of the demand, yet discounts are basically the same across products (0.61% to 0.64%). Discounts should be used to push slower moving brands and reward loyalty, while also protecting margins on high performing makes. Especially considering revenue is down (roughly 41% for the year), pricing needs to be intentional.

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.